

Hawkeye Audit Module — Gap Analysis + Roadmap

Hawkeye Go-To-Market Pack · 2026-04-29

Hawkeye Audit Module — Gap Analysis + Roadmap

Reference: 24-step expert process + remote-audit user flow + ICH Q7 / EU GMP Ch.9 / 21 CFR Part 11 + competitor analysis (Veeva · MasterControl · TrackWise · ETQ · Qualifyze · IQVIA SmartSolve · Greenlight Guru · AuditBoard · Workiva)

Last updated: 2026-04-29 Owner: Hawkeye Platform Audience: product · engineering · GTM Status: v1 — for review

§1 — TL;DR

The audit module covers ~60% of the 24-step expert flow with solid scaffolding for onboarding, scheduling, PAQ, deficiency, CAPA, and monitoring. **Five categories of gaps** cause the breakage you are seeing today and block enterprise pharma adoption:

1. **Intimation letter + supplier e-signature** — completely missing as a record type. No supplier acceptance ceremony before audit can proceed.
2. **Internal vs external auditor distinction + COI + availability scheduling** — model fields exist (`AvailabilityBlockModel` , `AuditorQualificationModel.coiDeclarations`) but the buyer's auditor dropdown does **not** filter by date availability or COI status, and the User schema doesn't model "in-house auditor" vs "third-party auditor".
3. **Auditor execution-questionnaire builder** — no UI for the auditor to compose a focused execution checklist by selecting categories/subcategories from a template; questions are auto-generated.
4. **Per-response live evidence (audio/video/photos/notes)** — `auditorAttachments` and `RemoteSessionModel` exist but are not wired into the execution UX; no video/screen-share provider; no real-time evidence streaming during a remote audit.
5. **Formal audit closure certification** — `phaseState` tracks CLOSURE but there is no certification artifact, no signed closure handoff, no QP-release-blocking signal on disqualification.

Market opportunity is real: only Qualifyze advertises a true supplier-side portal, no incumbent has shipped GenAI-drafted observations with citation-traceable evidence, no incumbent bundles video + screen-share + in-tool evidence annotation. **Hawkeye's wedge is a remote-audit cockpit + supplier-first portal at SMB-pharma price point** — the white-space between Qualifyze (per-audit marketplace) and Veeva (\$30K+ enterprise floor).

The roadmap below is **5 phases over ~5 calendar months**, with the first 2 phases unblocking the demo script and the remaining 3 closing market gaps.

§2 — Reference flow (what we are building toward)

Three reference flows merge into a single canonical model:

§2.1 The 24-step expert flow (auditable, Annex 16 compliant)

Source: `docs/04-processes/superuser-process-flow-24steps.md` . Steps 1-4 are PQ (already wired). Steps 5-24 are the audit-only scope this doc covers.

§2.2 The user's described "remote-audit" flow

Buyer sends intimation letter and/or PAQ → supplier signs intimation → supplier optionally fills PAQ → if PAQ not sent now, audit returns to buyer → buyer assigns auditor (internal or external) → auditor calendar updates with availability block → buyer's auditor dropdown filtered by date availability → auditor accepts → auditor sends PAQ + agenda → supplier fills PAQ + agenda → auditor sends execution questionnaire (selects questions/categories/subcategories) → auditor reviews + sends to supplier admin → supplier admin assigns sections to multiple users → each user fills questionnaire + attaches evidence → submits to auditor → auditor scans attachments + responses, captures notes/videos/photos/audio per response → prepares audit report → generates CAPAs

§2.3 ICH Q7 / EU GMP / 21 CFR Part 11 hard requirements

Requirement	Source	Hawkeye implication
Annual internal audit schedule covering all GMP scope	ICH Q7 §2.4 + EU GMP Ch.9	Audit program calendar with scope-coverage tracking
Risk-based supplier qualification + audit strategy	ICH Q7 §7 + EU GMP Ch.7	Supplier risk band drives audit cadence + checklist depth (Q9(R1) formality spectrum)
Quality Agreement between Contract Giver / Acceptor	EU GMP Ch.7	First-class QualityAgreement record, signed, audit-rights clause exposed
Audit report retained permanently, never deleted post-CAPA	EU GMP Ch.9	Append-only AuditReport collection; no destructive delete
Auditor independence from area being audited	EU GMP Ch.9 + ISO 19011	Cannot assign auditor with conflict-of-interest at supplier or area
21 CFR Part 11 audit trail (append-only, time-stamped, ≥ record retention)	21 CFR §11.10(e)	Already partially in place (AuditTrailService); needs lock-on-signature for AuditReport
E-signature with two distinct components, unique-per-individual	21 CFR §11.50, §11.200	Existing ElectronicSignatureModel covers this; needs to be extended to IntimationLetter recordType
QP-blockable supplier disqualification	EU GMP Annex 16	Supplier qualification status + audit-date provenance must surface to batch-release flow

§2.4 Canonical 26-step flow (24 expert steps + 2 reconciled additions)

Synthesized below. Step numbers prefixed S to disambiguate from expert #001-#024 .

#	Step	Owner	Triggers next
S01-S04	(PQ — already shipped)	Buyer · Supplier	Approved PQ
S05	Pre-Audit Communication / Intimation Letter	Buyer QA	Supplier signature
S05a	Internal vs External auditor decision	Buyer QA	Auditor pool filter
S06	Auditor COI declaration + Auditee acceptance	Auditor → Supplier	COI signed
S07	Audit Scheduling (calendar-aware)	Buyer QA	Mutually agreed dates
S07a	Auditor availability check + dropdown filter	System	Available auditors list
S08	Pre-Audit Documentation request (SMF, SOPs, specs)	Auditor	Docs uploaded
S09	Pre-Audit Questionnaire (PAQ)	Auditor → Supplier	PAQ submitted
S09a	Supplier admin assigns PAQ sections to teammates	Supplier admin	Sections assigned
S10	Audit Agenda	Auditor → Supplier	Agenda accepted
S10a	Auditor builds execution checklist from template	Auditor	Execution checklist signed off
S11	Audit Checklist standard verification	Auditor	Ready to execute
S12	Audit Execution (remote / on-site / hybrid)	Auditor + Supplier	Findings captured
S12a	Live remote evidence capture (video/screen-share/notes/photos)	Auditor + Supplier	Per-response media attached
S13	Opening meeting	Auditor + Buyer + Supplier	Minutes recorded
S14	Closing meeting reporting	Auditor → Supplier	Closing minutes signed
S15	Facility Certification (Approved / Conditional / Rejected)	Auditor + Buyer QA	Decision communicated
S16	Deficiency Reporting	Auditor → Supplier	Deficiencies categorized
S17	Deficiency Validation/Acceptance	Auditor + Supplier	Agreed list
S18	Audit Report	Auditor → Supplier	Report issued
S19	CAPA Plan	Supplier QA	CAPAs submitted
S20	Review & Acceptance of CAPA Plan	Auditor + Buyer QA	CAPAs accepted
S21	Audit Closure Certification	Auditor → Buyer	Closure signed
S22	CAPA tracking	Buyer QA	Effectiveness verified
S23	Supplier monitoring/rating	Buyer QA	Risk band updated
S24	Follow-up / requalification audit planning	Buyer QA	Next audit scheduled

The **bold rows** are the 5 reconciled additions that close the gaps between the 24-step expert flow and the user's described modern remote-audit reality.

\$3 — Current state mapping (per step)

Compiled from a code walk of `backend/src/routes/` , `backend/src/controllers/` , `backend/src/models/` , and `frontend/app/(console)/audits/` . Detail per step:

#	Step	Backend route	Frontend page	Status
S05	Pre-Audit Communication / Intimation Letter	auditPhaseRoutes.js (artifact upsert)	AuditArtifactDetail.tsx (renders intimation template)	■ Missing supplier signature ceremony — no recordType: "INTIMATION_LETTER" in ElectronicSignatureModel
S05a	Internal vs External auditor decision	—	—	■ No model field distinguishing internal employee vs external 3rd-party auditor on AuditorProfileModel
S06	Auditor COI declaration + acceptance	auditorQualificationRoutes.js	—	⚠ Model exists, no UI for buyer/supplier to view/accept COI
S07	Audit Scheduling	auditScheduleRoutes.js	/audits/[id]/summary	✅ Working
S07a	Auditor availability dropdown filter	auditorRoutes.js /availability GET/POST	—	⚠ AvailabilityBlockModel exists , AuditorSelector does NOT filter by date availability
S08	Pre-Audit Documentation request	documentControlRoutes.js (generic)	/document-control	⚠ Generic doc control, no audit-bound request
S09	Pre-Audit Questionnaire	preAuditRoutes.js + v2/questionnaires.js	/audits/[id]/questionnaire	✅ Working
S09a	Supplier admin assigns PAQ sections	questionnaireAssignmentRoutes.js	partially in /audits/[id]/questionnaire	⚠ Backend assignment model exists but only "one user per category"; no multi-user-per-section, no UI for supplier admin to bulk assign
S10	Audit Agenda	preAuditRoutes.js	/audits/[id]/summary	✅ Working
S10a	Auditor execution-checklist builder	auditorController.js createPreviewAuditQuestions	—	❌ No UI for auditor to select categories/subcategories and compose a focused execution checklist
S11	Audit Checklist standard verification	auditPhaseRoutes.js	/audits/[id]/template/management	✅ Working
S12	Audit Execution	auditPhaseController.js , auditorController.js	/audits/[id]/report (combined)	⚠ Functional but no dedicated execution-mode UI; auditor edits in report tab
S12a	Live remote evidence (video/screen-share/notes/photos/audio)	remoteAuditRoutes.js + RemoteSessionModel	—	❌ Model exists, no provider integration (Zoom/Teams/Jitsi), no live capture
S13	Opening meeting	—	—	■ No model, no UI
S14	Closing meeting reporting	auditPhaseController.js (notes)	—	⚠ Notes model exists, no formal closing workflow with sign-off
S15	Facility Certification	auditPhaseController.js (facilityOutcome field)	—	⚠ Field exists, no certification UI / signed artifact
S16	Deficiency Reporting	AuditReport.observations[]	/audits/[id]/report	✅ Working
S17	Deficiency Validation/Acceptance	auditRequestRoutes.js POST /:id/deficiency-validation	—	⚠ Endpoint exists, no UI
S18	Audit Report	auditorRoutes.js (report endpoints) + new /review	/audits/[id]/report	✅ Working
S19	CAPA Plan	capaV2Routes.js + capaRoutes.js	/buyer/capas + /supplier/capas	✅ Working (after CapaV2 mirror fix)
S20	Review & Acceptance of CAPA	capaV2Routes.js (status workflow)	/buyer/capas/[id]	✅ Working
S21	Audit Closure Certification	—	—	■ Phase advances to CLOSURE but no certification artifact

#	Step	Backend route	Frontend page	Status
S22	CAPA tracking	capaV2Routes.js (lastActivityAt , targetDate)	/supplier/capas	✅ Working
S23	Supplier monitoring/rating	supplierRiskRoutes.js + closure hooks	/buyer/suppliers/[id]/quality-events	⚠ Risk score exists, not visibly tied to audit history
S24	Follow-up / requalification audit planning	auditRequestsMasterModel.js auditType: 'RECERTIFICATION'	—	⚠ Field exists, no scheduler / reminder service

Coverage: 11 ✅ working · 11 ⚠ partial · 4 ❌ missing entirely · RemoteSession infrastructure inert.

§4 — Market study (competitor matrix + ICH Q7 implications)

§4.1 Competitor feature matrix (compressed)

Vendor	Auditor execution tools	Supplier portal	Internal vs external	Remote audit	Auditor availability	AI features	Price
Veeva Vault QMS	Audit Program Planning, Auditor Qualification	External Response Collaboration auto-provisions accounts	Both (modeled)	None native	Auditor qualification gating; no scheduler	Vault AI being added	\$25M ~\$62,400 · large pharma
MasterControl Audit	Wizard-driven lifecycle, supplier scorecard tied to risk	Limited internal-user oriented	Both (FDA, ISO, internal/external types)	None native	None native	Some AI in newer modules	Quoted
TrackWise (Sparta/Honeywell)	Workflow engine; "400+ audits/yr"	Limited	Internal, third-party, regulatory	None	None	TrackWise AI / QualityWise-AI (event triage)	Quoted
ETQ Reliance	Plan→checklist→execute→CAPA · mobile any-device	Supplier perf tracking, no full portal	Both	Mobile = field-friendly	None	Some smart recommendations	Mid-ent
Qualifyze	Marketplace — outsource the audit; 5,000+ shared reports	Yes — auditee portal (only competitor with this)	External-supplier-focused	Yes — explicit remote audits via video conf + doc exchange	AI-powered scheduling	"AI-powered platform · 65% faster · 50% cost saving"	Per-price
IQVIA SmartSolve	Mobile audit + tight CAPA integration	Across 20+ life-sciences modules	Both	Mobile-friendly	None	Generic ML	Quoted
Greenlight Guru	Internal/external/supplier/regulatory audits	Supplier mgmt module	Both (medical-device-centric)	None	None	None visible	Mid-SaaS
AuditBoard	Strongest UX in space; 200+ integrations	None pharma-specific	Cross-industry	None	None	Strong reporting/automation	\$30M typical
Workiva Wdesk	Risk assessment, scoping, evidence collection	None	Internal-audit-focused (SOX origin)	None	None	Workiva Gen AI assistant '25	Enterprise quote

§4.2 White-space (where nobody has shipped)

- Supplier-first portal as a first-class user experience** (sign intimation, fill PAQ, assign sections, respond to CAPA in their own UI) — Qualifyze partial; everyone else absent.
- Integrated remote-audit cockpit** — video + screen-share + in-tool evidence annotation as one product, not Zoom + email + portal.
- GenAI-drafted observations with citation-traceable evidence** — the PDA Letter pattern (summarization + clickable source traces + human approval). Incumbents talk AI but ship triage/scheduling.
- Auditor availability + COI + qualification dropdown filter** — undermarketed by everyone.
- Risk-based formality (ICH Q9(R1) "formality spectrum")** as a first-class UX concept — high-risk supplier visibly gets deeper checklist than low-risk.

§4.3 ICH Q7 / regulatory pressure shaping the roadmap

- **EU GMP Chapter 9** = audit reports never deletable post-CAPA closure → append-only `AuditReport` collection, lock-on-signature.
- **EU GMP Chapter 7** = Quality Agreement is mandatory between Contract Giver/Acceptor → add `QualityAgreement` first-class record with audit-rights clause exposed.
- **EU GMP Annex 16** = QP must confirm supplier audits within prior 3 years → expose **supplier qualification status with audit-date provenance** to batch-release flow; surface **disqualification** → **batch-block signal**.
- **21 CFR Part 11** = e-signature with two components, unique per individual, never reused → already in place but **must extend to `INTIMATION_LETTER` recordType**.
- **ICH Q9(R1)** (Step 4 in 2023, amended 2025) = "formality spectrum" — high-risk audits visibly more formal → **risk-band-driven checklist depth** must be in the UX.
- **2024-2025 FDA inspection trends** — data integrity 79% of warning letters, CMO oversight a 2025 enforcement focus, foreign sites 33% of warning letters → ship strong **data-integrity question pack** + **CMO oversight pack** as defaults.

§5 — Gap analysis (top 12, prioritized)

P0 = blocks the demo script today · P1 = blocks first-customer pilot · P2 = blocks enterprise sale · P3 = nice to have

#	Gap	Priority	Where it bites
G1	No supplier e-signature on intimation letter (S05)	P0	Demo step 5 fails; supplier just acknowledges, doesn't sign
G2	Buyer auditor dropdown not filtered by availability + COI (S05a/S07a)	P0	User reports buyer can pick unavailable auditor → broken assignment
G3	Internal vs external auditor not modeled	P0	User explicitly asked for this; no UI exists today
G4	Supplier admin can't assign PAQ sections to multiple teammates (S09a)	P0	Demo UC-5 demands it; backend half-built
G5	Auditor execution-checklist builder MIA (S10a)	P0	Auditor can't compose focused checklist; demo UC-6 papered over
G6	Audit-only flow has multiple navigation/state bugs	P0	"Phase closed" errors, stale auditor placeholder, broken CAPA mirror — partly addressed already
G7	No live remote evidence capture (audio/video/screen-share/notes/photos per response) (S12a)	P1	Differentiator vs incumbents; user explicitly asked
G8	No formal closure certification artifact (S21)	P1	Audit advances through CLOSURE without a signed handoff
G9	Internal audit program calendar (annual schedule with scope coverage) missing	P1	ICH Q7 §2.4 + EU GMP Ch.9 mandatory; needed for self-inspection use case
G10	Quality Agreement record + audit-rights clause	P2	EU GMP Ch.7 mandatory for supplier qualification; QP needs it for Annex 16
G11	Risk-band-driven checklist depth (Q9(R1) formality spectrum)	P2	Differentiator + 2025 FDA expectation
G12	AI observation drafter with citation-traceable evidence	P2	Market white-space; PDA Letter pattern

§6 — Roadmap (5 phases, ~5 calendar months)

Each phase ends with a **demoable + sellable** milestone. Backend-first within each phase, then frontend, then polish.

Phase 1 — Stabilize the audit-only happy path (3-4 weeks)

Theme: make the demo script work end-to-end without surprises. Hits gaps **G1, G2, G3, G4, G5, G6**.

- **G3 → User schema:** add `auditorAffiliation: enum('internal', 'external')` and `auditorOrgId` to `AuditorProfileModel`.
- **G2/G7a → Availability filter:** `AuditorSelector` component calls a new `GET /api/auditors/available?supplierProposedDates=...&durationDays=...&auditorAffiliation=...` that joins `AuditorQualification.coiDeclarations` × `AvailabilityBlockModel` and returns only auditors with no conflict + no COI block. Buyer dropdown re-renders from this.
- **G1 → Intimation letter signature:** extend `ElectronicSignatureModel.recordType` enum with `INTIMATION_LETTER`; add `POST /api/audits/:id/intimation/sign` endpoint; add a "Sign intimation letter" CTA on the supplier-side audit detail page; lock the artifact `data` on signature with `signatureId` reference.
- **G4 → Multi-user PAQ section assignment:** extend `QuestionnaireSectionAssignmentModel` to support `assignedToUserIds[]` (array, not single); add a "Assign sections" page for supplier admin where they bulk-assign categories to teammates; section-level filter on `/audits/[id]/questionnaire` so each assignee sees only their sections.
- **G5 → Execution-checklist builder:** new auditor page `/audits/[id]/execution/builder` that lists all available audit-question categories from the active template, lets the auditor check/uncheck and reorder, then writes the curated set into `AuditQuestions[]` with `inExecutionScope: true`. The execution UI then filters by this flag.

- **G6 → Audit-only navigation/state:** continuation of the previous bug-fix batches — verify HK-0000086, HK-0000087 flows work with the buyer-tenant-scope query already shipped.

Exit criteria: every step S05-S20 in the canonical 26-step flow has a working UI and backend; one persona walkthrough run captures all of them with no empty screenshots.

Phase 2 — Closure + audit program (3-4 weeks)

Theme: close the audit cleanly and surface the loop back to the buyer's supplier program. Hits gaps **G8, G9, G10, G11**.

- **G8 → Closure certification:** new `AuditClosureCertificate` model with `signatureId` reference; `POST /api/audits/:id/close-certificate`; PDF rendered with template; QP-blocking signal exposed via `audit.closureOutcome` and joined into `SupplierRiskSnapshot`.
- **G9 → Audit program calendar:** new `AuditProgramModel` with annual schedule + GMP scope coverage matrix; calendar UI for buyer QA showing planned vs completed per scope area (premises / equipment / personnel / docs / production / QC / distribution / complaints / recalls / self-inspection).
- **G10 → Quality Agreement:** new `QualityAgreementModel`; signed contract record; audit-rights clause exposed as a queryable field; surface as a tab on `/buyer/suppliers/[id]`.
- **G11 → Q9(R1) formality spectrum:** add `riskBand` resolution to template selection so high-risk supplier auto-loads a deeper checklist; surface "Q9 formality: HIGH" badge on the audit detail.

Exit criteria: EU GMP Chapter 7+9+Annex 16 inspection-readiness checklist passes manual review; closure certificate renders and is non-deletable; audit program calendar shows annual scope coverage with remaining-scope tracking.

Phase 3 — Remote-audit cockpit (4-6 weeks)

Theme: ship the integrated remote-audit experience that no incumbent has bundled. Hits gap **G7**.

- **Provider integration:** Jitsi Meet self-hosted or Daily.co API (cheaper than Zoom/Teams enterprise) wired into `RemoteSessionModel`; meeting URLs + session tokens minted server-side.
- **Auditor cockpit page:** `/audits/[id]/execution/remote` — left rail with the curated execution checklist, center pane with the embedded video + supplier shared screen, right rail with per-question evidence dock that captures audio/video/photos/notes timestamped to the question.
- **Supplier remote-audit page:** `/supplier/audits/[id]/remote` — joins the same session; can share screen + upload evidence in real time; sees the question the auditor is currently on.
- **Per-response media:** `AuditQuestion.responseMedia[]` array with `{type, mediaRef, capturedAt, capturedBy}`; uploads go through S3 with WebSocket progress; large file support (already 25MB JSON limit; raise S3 multipart for >100MB videos).
- **Recording + retention:** session recording stored in `RemoteSessionModel.recordingRef`; lifecycle policy 7 years (matches batch retention); accessible only to auditor + buyer QA + supplier admin.

Exit criteria: an auditor and a supplier can run a full remote audit on a single screen each, end-to-end, with every observation timestamped to the moment in the recording.

Phase 4 — AI observation + report drafter (4-5 weeks)

Theme: ship the GenAI capability nobody has — citation-traceable observation drafting. Hits gap **G12**.

- **Source-of-truth corpus:** ingest the audit's curated checklist + supplier PAQ responses + uploaded evidence + ICH Q7 / EU GMP / 21 CFR cited clauses into a vector store keyed by audit ID.
- **Observation drafter agent:** for each finding the auditor flags, the agent returns a draft observation with **inline citations** ([1], [2]) clickable back to source paragraph in evidence or guidance. Pattern follows PDA Letter: summarization + citations + human approval.
- **Severity + GMP-classification suggester:** based on cited clause + severity heuristics, agent suggests Critical / Major / Minor + suggested CAPA target date; auditor reviews and accepts or overrides.
- **Report assembler:** existing endpoint upgraded to assemble report sections from observations with citations preserved; report PDF includes a "Sources" appendix.
- **Audit-trail:** every AI suggestion logged in `agent-usage-events` with input + output + acceptance/edit decision (already in place).

Exit criteria: a real audit ends in a drafted report whose every observation has a clickable citation; auditor edit time decreases >50% measured against baseline; agent-usage cost-per-audit visible in the AI ROI dashboard.

Phase 5 — Internal audit + program maturity (3-4 weeks)

Theme: open the second use-case — internal audit / self-inspection — that doubles addressable market. Hits residual gaps + ICH Q7 §2.4.

- **Internal audit mode:** audit type already has enum value; surface it in the New Audit form; internal audits use the buyer's own employees as auditors (filter `auditorAffiliation === 'internal'`) auditing a buyer site rather than a supplier.
- **Scope-coverage matrix:** Audit Program calendar (Phase 2) extended to track which GMP scope areas have been covered in the current cycle; flag uncovered areas as risk.
- **Management review pack:** auto-generated quarterly summary of audit findings + CAPA closure rates + supplier risk movement, ready for management review per ICH Q10.
- **Requalification audit auto-scheduler:** `auditType: 'RECERTIFICATION'` triggers a calendar invite + email reminder N days before the prior audit's `validUntil`.
- **Public-source audit-readiness dashboard:** for the buyer's own sites, surface FDA 483 / warning letter trends affecting their region/product class and pre-load the relevant question packs.

Exit criteria: an internal QA team can run their full annual self-inspection program through Hawkeye; management review pack auto-generates from real data; recertification reminders fire on schedule.

§7 — Recommendations + immediate-next-steps

Cut from scope (for now):

- Building our own video provider — use Jitsi/Daily.co.
- Mobile-native app — desktop browser is sufficient for v1; ETQ's mobile UX is their differentiator, not ours.
- A marketplace/shared-audit model like Qualifyze — different business model, not our wedge.

Double down on (for differentiation):

- Supplier-first portal experience (existing investment; finish multi-user section assignment in Phase 1).
- Citation-traceable AI observation drafting (Phase 4).
- Integrated remote-audit cockpit (Phase 3).
- Risk-based formality (Q9(R1)) as a visible UX concept (Phase 2/4).

Ship next, in order:

1. Stop bug-fix sprint; start Phase 1 with the 6 gaps above (G1-G6).
2. Update `07-pharma-demo-script.md` so its menu paths reference the section-labelled top-bar (already shipped) and add a UC for the new intimation-letter signature ceremony (Phase 1 deliverable).
3. Run the `bugfix-e2e-suite.test.mjs` after every Phase 1 backend change to catch regressions.
4. After Phase 1 lands, add a Phase-2 "Internal Audit Program" demo persona (Sanjay — Buyer QA Director) with a separate UC track.

Sales / GTM signals to instrument:

- Per-tenant audit completion time (target: <60% of baseline by end of Phase 4).
 - Per-audit AI agent cost (track in AI ROI dashboard; target <\$5/audit).
 - Auditor finding acceptance rate (target: >80% of AI-suggested observations accepted with minor edit).
 - Supplier intimation-signature time-to-sign (target: <72hrs median).
-

§8 — Open questions for product

1. **Internal-vs-external auditor org modeling** — should an internal auditor be tied to a `BuyerOrg.employees[]` or a separate `InternalAuditorTeam` record? Affects how COI is computed.
 2. **Quality Agreement** — should the QA template be tenant-configurable or platform-default? FDA + EU expectations differ slightly.
 3. **Recording retention duration** — 7 years matches batch retention but storage cost is non-trivial; tenants may want to override (down to 2-3 years for low-risk suppliers).
 4. **AI provider strategy** — keep Gemini Flash-Lite + Claude Sonnet for Phase 4 drafting, or add Anthropic-only for higher fidelity at the cost? Open until pricing data lands.
 5. **Mobile** — defer entirely or PWA in Phase 5? ETQ's pharma customers consider mobile near-mandatory for plant-floor on-site audits.
-

§9 — Sources

Regulatory:

- ICH Q7 Guideline (database.ich.org)
- ICH Q9(R1) Step 4 (Federal Register, 2023; correction Jan 2025)
- EU GMP Chapter 7, Chapter 9, Annex 16 (ECA Academy, gmp-compliance.org)
- 21 CFR Part 11 (eCFR)
- ICH Q7 Q&As (database.ich.org)

Competitor product pages: Veeva Vault QMS · MasterControl Audit · Sparta TrackWise · ETQ Reliance · Qualifyze · IQVIA SmartSolve · Greenlight Guru · AuditBoard · Workiva Wdesk

Industry direction: PDA Letter (AI in pharma audit readiness), Pharma Manufacturing (hybrid inspection model), TCS (remote audit automation), Leucine (2025 FDA Warning Letter trends), Pharmaphorum (controlling AI hallucinations), Clinical Leader (FDA AI guidance)

Hawkeye internal docs:

- [docs/04-processes/superuser-process-flow-24steps.md](#)
 - [docs/04-processes/gmp-audit-data-flow.md](#)
 - [docs/05-feature-guides/audit-only-feature-guide.pdf](#)
 - [docs/06-go-to-market/07-pharma-demo-script.md](#)
-